```python
import sqlite3

try:
    # Connect to SQLite Database
    sqliteConnection = sqlite3.connect("sql.db")
    cursor = sqliteConnection.cursor()

    print("Database Connected Successfully")

    # SQLite Version
    cursor.execute("SELECT sqlite_version();")
    version = cursor.fetchone()
    print("SQLite Version:", version[0])

    # Create Table
    cursor.execute("""
    CREATE TABLE IF NOT EXISTS Student(
        id INTEGER PRIMARY KEY,
        name TEXT,
        age INTEGER,
        department TEXT
    )
    """)
    print("Table Created")

    # Delete old data (optional)
    cursor.execute("DELETE FROM Student")
```

```
# Insert Data
students = [
    (1, "Dinesh", 21, "CSE"),
    (2, "Rahul", 22, "ECE"),
    (3, "Priya", 20, "IT"),
    (4, "Kumar", 23, "CSE"),
    (5, "Anitha", 21, "AIDS")
]

cursor.executemany("INSERT INTO Student VALUES (?,?,?,?)", students)
sqliteConnection.commit()
print("Records Inserted")

# Select Data
print("\nAll Records")
cursor.execute("SELECT * FROM Student")
for row in cursor.fetchall():
    print(row)

# WHERE Clause
print("\nWHERE department='CSE'")
cursor.execute("SELECT * FROM Student WHERE department='CSE'")
for row in cursor.fetchall():
    print(row)

# ORDER BY Clause
print("\nORDER BY age")
cursor.execute("SELECT * FROM Student ORDER BY age")
```

```
for row in cursor.fetchall():
    print(row)

# LIMIT Clause
print("\nLIMIT 3")
cursor.execute("SELECT * FROM Student LIMIT 3")
for row in cursor.fetchall():
    print(row)

# Update Data
cursor.execute("UPDATE Student SET age=24 WHERE id=1")
sqliteConnection.commit()
print("\nAge Updated")

# Update Specific Column
cursor.execute("UPDATE Student SET department='AI&DS' WHERE id=2")
sqliteConnection.commit()
print("Department Updated")

# Display Updated Records
print("\nUpdated Records")
cursor.execute("SELECT * FROM Student")
for row in cursor.fetchall():
    print(row)

# Delete Data
cursor.execute("DELETE FROM Student WHERE id=5")
sqliteConnection.commit()
```

```
    print("\nRecord Deleted")

    print("\nAfter Delete")
    cursor.execute("SELECT * FROM Student")
    for row in cursor.fetchall():
        print(row)

    # Drop Table
    cursor.execute("DROP TABLE IF EXISTS Student")
    sqliteConnection.commit()
    print("\nTable Dropped")

except sqlite3.Error as error:
    print("SQLite Error:", error)

finally:
    if sqliteConnection:
        cursor.close()
        sqliteConnection.close()
        print("\nSQLite Connection Closed")
```

ANSWER:

Database Connected Successfully

SQLite Version: 3.40.1

Table Created

Records Inserted

All Records

(1, 'Dinesh', 21, 'CSE')

(2, 'Rahul', 22, 'ECE')

(3, 'Priya', 20, 'IT')

(4, 'Kumar', 23, 'CSE')

(5, 'Anitha', 21, 'AIDS')

WHERE department='CSE'

(1, 'Dinesh', 21, 'CSE')

(4, 'Kumar', 23, 'CSE')

ORDER BY age

(3, 'Priya', 20, 'IT')

(1, 'Dinesh', 21, 'CSE')

(5, 'Anitha', 21, 'AIDS')

(2, 'Rahul', 22, 'ECE')

(4, 'Kumar', 23, 'CSE')

LIMIT 3

(1, 'Dinesh', 21, 'CSE')

(2, 'Rahul', 22, 'ECE')

(3, 'Priya', 20, 'IT')

Age Updated

Department Updated

Updated Records

(1, 'Dinesh', 24, 'CSE')

(2, 'Rahul', 22, 'AI&DS')

(3, 'Priya', 20, 'IT')

(4, 'Kumar', 23, 'CSE')

(5, 'Anitha', 21, 'AIDS')

Record Deleted

After Delete

(1, 'Dinesh', 24, 'CSE')

(2, 'Rahul', 22, 'AI&DS')

(3, 'Priya', 20, 'IT')

(4, 'Kumar', 23, 'CSE')

Table Dropped

SQLite Connection Closed

============ RESTART: E:/query lab/CLAS WORK SQL CONNECT TO IDEL.py ============

Database Connected Successfully

SQLite Version: 3.40.1

Table Created

Records Inserted

All Records

(1, 'Dinesh', 21, 'CSE')

(2, 'Rahul', 22, 'ECE')

(3, 'Priya', 20, 'IT')

(4, 'Kumar', 23, 'CSE')

(5, 'Anitha', 21, 'AIDS')

WHERE department='CSE'

(1, 'Dinesh', 21, 'CSE')

(4, 'Kumar', 23, 'CSE')

ORDER BY age

(3, 'Priya', 20, 'IT')

(1, 'Dinesh', 21, 'CSE')

(5, 'Anitha', 21, 'AIDS')

(2, 'Rahul', 22, 'ECE')

(4, 'Kumar', 23, 'CSE')

LIMIT 3

(1, 'Dinesh', 21, 'CSE')

(2, 'Rahul', 22, 'ECE')

(3, 'Priya', 20, 'IT')

Age Updated

Department Updated

Updated Records

(1, 'Dinesh', 24, 'CSE')

(2, 'Rahul', 22, 'AI&DS')

(3, 'Priya', 20, 'IT')

(4, 'Kumar', 23, 'CSE')

(5, 'Anitha', 21, 'AIDS')

Record Deleted

After Delete

(1, 'Dinesh', 24, 'CSE')

(2, 'Rahul', 22, 'AI&DS')

(3, 'Priya', 20, 'IT')

(4, 'Kumar', 23, 'CSE')

Table Dropped

SQLite Connection Closed